Exercise 2

Using

$$k_B = 8.617 \times 10^{-5} \ \mathrm{eV/K},$$

calculate $k_B T$ for

$$T = 1 \ \mathrm{K}, \qquad T = 10^{-2} \ \mathrm{K}, \qquad T = 10^{-3} \ \mathrm{K}\,.$$

Comment on why millikelvin temperatures are needed for nuclear polarization.

Exercise 3

The Bohr magneton and the nuclear magneton are

$$\mu_B = 5.79 \times 10^{-5}\ \mathrm{eV/T}, \qquad \mu_N = 3.15 \times 10^{-8}\ \mathrm{eV/T}\,.$$

- Calculate $\mu_B B$ and $\mu_N B$ for $B = 1\ \mathrm{T}$.
- Find the ratio $\dfrac{\mu_B}{\mu_N}\cdot$
- Compare both energies with $k_B T$ at $T = 1\ \mathrm{K}$.

Explain why electron spins are much easier to polarize than nuclear spins.